

STATEMENT OF INTEREST AND RELEVANT EXPERIENCE FOR SCHOOL OF COMPUTING AT KAIST

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My research lies at the intersection of 3D vision, multimodal understanding, and spatial reasoning. I am broadly interested in building systems that jointly reason over geometry, appearance, and language - systems capable of understanding not merely what a scene looks like, but what it contains, how it is structured, and how it relates to the world described in natural language. 3D Gaussian Splatting has emerged as a particularly compelling representation for this kind of understanding: it is expressive, differentiable, and increasingly amenable to semantic manipulation. My goal, as I look toward graduate study, is to push GS-based representations beyond novel view synthesis toward richer scene understanding - enabling object-level reasoning, part-level retrieval, and language-grounded spatial analysis in dynamic, real-world environments.

I am currently a third-year undergraduate in the IT Talent Programme at the University of Science, VNU-HCM (HCMUS) - admitted among the top 45 students selected faculty-wide by competitive examination - with a GPA of 3.94/4.0 and expected graduation in May 2027. KAIST School of Computing is among the institutions I am most seriously considering for graduate study. I am applying to the Global Preview Week 2026 as a means of engaging with the research community at KAIST in person, and of grounding that consideration in direct intellectual exchange with faculty and graduate students.

Research Experience

My research has developed across two settings. Since June 2024, I have been a research student at the Software Engineering Lab (SELab) at HCMUS, under the supervision of Prof. Minh Triet Tran. In early 2026, I was additionally selected into one of ten inaugural research groups at Saigon AI Hub (SAIH) - Vietnam's first open AI research hub and the first organisation worldwide chosen for Google Labs' AI Futures Fund - where I work on multimodal sport reasoning and analysis.

My primary research interest is **3D spatial intelligence**, and specifically the question of how Gaussian Splatting representations can be made semantically aware - capable not merely of rendering scenes, but of understanding and reasoning over them. My earliest work in this direction was a contribution to a part-level shape retrieval framework for GS-format 3D objects (SHREC 2025), addressing similarity matching within this relatively novel representation space, published in *Computers & Graphics* (Elsevier, Q2). This experience gave me a concrete appreciation of the core challenge: performing semantics-aware reasoning directly within GS representations, which I regard as foundational to any system that aims to understand, rather than merely reconstruct, a 3D scene. It is this problem - bridging geometry, appearance, and language within a GS framework - that defines my research direction going forward.

Alongside this, I have led work in multimodal content generation and understanding. I developed an end-to-end pipeline for automated news thumbnail generation, combining LLM-based semantic prompt refinement with diffusion-model image synthesis, and introduced a dual-filtering strategy that selectively suppresses noisy visual cues to improve generation quality - contributing as first author to two publications (SOICT 2025; MediaEval 2025 Workshop). I also led work on metric-scale athlete localisation from single broadcast frames, proposing Boundary-Aware Adaptive Tiling for semantics-guided sliced inference and combining it with camera-calibrated ray casting to lift 2D detections to world coordinates, achieving **1st place** on the SoccerNet 2026 Spiideo SynLoc Challenge (97.67 mAP) and accepted at ICMV 2026 as first author.

Further contributions include TreeSoc, a structured hierarchical reasoning framework for soccer video QA achieving state-of-the-art on SoccerBench (ICMV 2026); SnapSeek 4.0, an event-centered lifelog retrieval system with hierarchical multimodal captioning and segmentation-based regional embeddings (ACM LSC @ ICMR 2026); and GUARD, a tri-modal deepfake forensics framework achieving top 3 on the ACM MM 2026 TRIDENT Grand Challenge video track (under review, ACM MM 2026, CORE A*). Across these projects, I have co-authored seven research outputs spanning 3D vision, video understanding, multimodal generation, and forensics.

Even when spanning diverse domains - video understanding, multimodal generation, and forensics - my core interest has remained constant: advancing the reasoning capabilities of models that operate over 3D

Gaussian Splatting representations. Where most GS research focuses on reconstruction fidelity, I am drawn to the harder problem of semantic understanding - building systems that can parse, retrieve, and reason over what a Gaussian scene contains, not just how it looks.

Research Vision and Graduate Goals

The problem I want to work on at the graduate level is multimodal 3D scene understanding: enabling machines to parse a scene not only in terms of its geometry and appearance, but in terms of the objects, parts, and relationships it contains, and to communicate and reason about these in natural language. I see 3D Gaussian Splatting as a particularly promising substrate for this, precisely because it is an explicit, object-composable representation that supports both rendering and, increasingly, semantic manipulation. The open questions I find most compelling concern how to ground language in GS representations - how to identify, retrieve, and reason about parts and objects within a scene without reducing the problem to 2D projection - and how to do this in settings that are dynamic rather than static.

My longer-term trajectory points toward a PhD and, beyond that, toward research that bridges 3D scene understanding and embodied AI - systems that must not only perceive but act within and reason about a physical world.

Fit with KAIST School of Computing

Two research groups at KAIST SoC are directly relevant to the direction I want to pursue.

The **KAIST Visual AI Group**, led by Associate Professor Minhyuk Sung, works on problems that sit at the core of my research agenda. Prof. Sung's group has produced work on optimisation-based editing of GS representations, proxy-free Gaussian splat deformation, and part-level generative modelling with BoxS-plitGen - each of which engages directly with the challenge of doing semantics-aware reasoning within GS scenes rather than treating them as opaque rendering outputs. His group's concurrent work on multimodal spatial intelligence and perspective-aware reasoning in vision-language models connects to the language-grounding dimension of my interest. The intellectual overlap between my GS-based part retrieval experience and the questions his group is pursuing is, I believe, substantive rather than superficial.

The **Computer Vision and Learning Lab (KCVL)**, led by Professor Tae-Kyun Kim, approaches the problem of 3D understanding from a complementary angle. Prof. Kim's work on physics-based generative AI - including MPMAvatar, which uses Gaussian Splatting combined with physics simulation to model dynamic 3D avatars - and his broader research on 3D human reconstruction and activity recognition in video, engages with a dimension of scene understanding I find compelling: the need to model not just appearance but the physical processes that govern how objects move and interact. His emphasis on grounding visual AI in physical reality connects naturally to my interest in understanding dynamic scenes, and to the sports video analysis work I have been conducting.

Closing

I am ranked in the top 2% of my faculty, with Dean's Honor List distinctions in 2024 and 2025, and I am fully available for the entire program period of September 6-12, 2026. I am at a stage in my development where visiting KAIST and engaging directly with its research community would be genuinely consequential for the decisions I am making about graduate study. I am grateful for KAIST's support of this programme and look forward to the possibility of being part of it.

Sincerely,



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Place: Ho Chi Minh City, Vietnam